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Enhancing Internet of Things security using performance gradient boosting for network intrusion detection systems

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ABSTRACT

The rise of the Internet of things (IoT) changes inter-device communication to increase efficacy and livability conditions in all sectors. However, it is proportion of security vulnerabilities making IoT networks prime targets for sophisticated cyber threats. The deployment of solution methods is needed to embark on new grounds in exploring IoT cybersecurity with the current state-of-the-art machine learning (ML) techniques to fortify IoT networks from such malicious threats. This paper highlights critical vulnerabilities in IoT traffic through a detailed analysis and performance evaluation of the state-of-the-art ensemble classifiers, eXtreme gradient boosting (XGBoost) and light gradient boosting machine (LGBM), to understand their detection capability for a diverse set of cyber-attacks. From the results, it can be seen that XGBoost and LGBM classifiers outperformed the conventional models with an extraordinary average accuracy of 99.553 % and 99.651 %, respectively in the definition of true threats. The performance metrics proved better detection capabilities of the classifiers with the potential that their use affords to minimize false positives and false negatives, which are preponderant considerations for the integrity of an IoT network. Further comparative analysis tries to furnish the strengths and limitations of these classifiers and propose a practical framework for their implementation to strengthen real-life IoT environments against cyber threats. This work delivers in-depth findings on IoT network behaviors and threat patterns using synthetic minority oversampling technique (SMOTE) for class balancing and analyzing the importance of features in IoT behavioral RT-IoT2022 dataset. This is also considered to be a foundational resource for creating cutting-edge AI-driven defense mechanisms to tackle ever-evolving cybersecurity threats.

1. Introduction

In today's digital age, the Internet of things (IoT) revolutionizes connectivity by bringing together a wide range of physical objects. IoT is a network that globally contains many connected devices that depend on networks, sensors, communication and processed information. IoT will enable the companies to improve their production and optimize their competitive power in the world of business [1]. More than just a technological improvement, this innovation represents a new era of human lifestyle, efficiency, and interaction with the physical world. IoT integrates virtually all fields of life and industry, from smart homes that enhance living comfort and security to wearables that monitor health in real time and sensors in agriculture that enable optimal farming

practices [2]. These applications will contribute significantly to advancements in vital sectors like healthcare, manufacturing, and urban development, leading us toward a more efficient and connected future. IoT systems can collect and exchange information by using mobile phones or computers. It has been confirmed that there are crucial threats when using IoT which demand strong security procedures [3]. The rapid growth of IoT devices across various sectors offers many benefits but poses significant security challenges. One major challenge is the inherent vulnerability of the IoT ecosystem, which comprises devices ranging from those with no computational power to highly advanced ones. Given the extensive distribution of devices and the heterogeneity in protocols and standards, ensuring data confidentiality, integrity, and availability is a complex task [4,5]. Addressing these challenges requires

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a multifaceted approach that carefully considers all aspects of IoT security [6].

The IoT revolution has brought breakthrough innovations in smart homes, wearables, and linked vehicles, enhancing traditional operational efficiencies and bringing the quality of life to its maximum depth. However, the number of IoT devices is expected to grow exponentially, and this growth has highlighted some complex security challenges that pose a fundamental threat to the confidentiality, integrity, and availability of information over these networks [7]. IoT networks mainly consist of devices with limited computational capabilities and low security. These devices' considerable diversity and distribution make it difficult to formulate any universal solution for securing them, leaving such networks as prime targets for cybercriminal exploits. In modern days, this trend has been continuing with scary developments in the sophistication and occurrence of cyber-attacks on IoT devices, besides also observing an exponential surge in malware, ransomware, and bot-net activities designed to compromise these vulnerabilities [8]. Such deteriorating cybersecurity challenges have also led to the most development focused on embedding machine learning (ML) and artificial intelligence (AI) tactics to strengthen IoT security. With these advanced methodologies, network traffic pattern analysis detects and processes cyber threats effectively, showing a more dynamic approach to real-time threat detection and mitigation. The success of the ML-based intrusion detection systems (IDSs) will thus depend on the comprehensiveness of datasets that represent details of complexities and diversities faced in real-life scenarios of IoT network traffic [9]. A key development is the RT-IoT2022 dataset [10], which represents the largest collection of IoT devices and attack scenarios. This dataset will be an enabler for designing and evaluating ML-based IDS. It will also serve as an indispensable dataset for researchers and cybersecurity professionals considering the behavior study within IoT networks and their threat models. The RT-IoT2022 dataset will simultaneously emulate benign and malicious activity over the network and form a basis for devising a real-time detection and response mechanism of threats in an IoT environment.

This work addresses the pressing need for advanced security solutions to safeguard IoT networks from a continually growing range of cyber threats. It builds upon the description of the RT-IoT2022 dataset and has the following contribution objectives:

- Identify specific attributes and vulnerabilities associated with various cyber-attacks targeting IoT devices, focusing on the characteristics of attacks such as Brute-Force SSH, DDoS, and Nmap scanning, within the context of diverse IoT devices.
- Assess the effectiveness of recent ensemble classifiers, particularly XGBoost and LGBM, the RT-IoT2022 dataset, comparing their performance against relevant baseline methods.
- Identify the effectiveness of XGBoost and LGBM in detecting different attack types, handling data complexity, and managing computational resources, providing practical insights for cybersecurity professionals.
- Propose an actionable strategy addressing specific challenges related to data collection, model training, and integration with existing infrastructure, including recommendations for mitigating data drift, resource constraints, and scalability issues.

Through a detailed exploratory data analysis of the RT-IoT2022 dataset, this study provides invaluable insights into the dynamics of IoT network traffic and the evolving landscape of cyber threats. Furthermore, by benchmarking the performance of advanced ensemble classifiers and validating their efficacy through rigorous statistical analysis, this research contributes significantly to developing robust and adaptive IoT security frameworks. This paper sets the stage for an in-depth exploration of IoT security, highlighting the pivotal role of innovative ML-based solutions in protecting interconnected devices against sophisticated cyber threats.

The rest of this article is structured as follows. Section 2 discusses the evolution of IoT technologies and the emerging complexity of cybersecurity threats, highlighting the role of the RT-IoT2022 dataset and recent strides in optimizing AI algorithms for resource constrained IoT devices. Section 3 describes the methodological approach, focusing on the use of ensemble classifiers to detect cyber threats within IoT networks and describing the pivotal role of the RT-IoT2022 dataset, the used data balancing strategies, including the use of SMOTE, and feature importance analysis. Section 4 introduces the experimental design and setup, providing a blueprint for replicability and scrutiny, covering dataset characteristics, preprocessing techniques, algorithm selection rationale, and evaluation metrics. Section 5 presents the results and offers a critical interpretation of the performance of various classifiers including a rigorous statistical analysis. Section 6 discusses the results and evaluation of the tested models. Finally, conclusions as well as future work of this study are presented in Section 7.

2. Related works

Technology has developed, and the nature and variety of threats in cyberspace have evolved, too, and intrusion detection requires sophisticated solutions. The RT-IoT2022 dataset is a rich collection of data derived from devices through IoT and methodologies of network attacks. This section discusses the developments using this dataset and the improvement of IDS due to the vulnerability of IoT. As part of their contributions, several works highlight the need to optimize AI algorithms to suit resource-constrained IoT devices. A few of the most noted developments include quantized AutoEncoder (QAE) models, which are meant to handle computational constraints and would be enlightened through sparsity-based pruning, clustering, and quantization techniques. These models were validated over the RT-IoT2022 dataset and displayed competitive effectiveness in anomaly detection while inflicting minimum carrying overhead. ML and deep learning (DL) are the two other tracks through which security can be enhanced in IoT. In this regard, such supervised learning techniques are applied for authentication and intrusion detection that use features like the received signal strength indicator (RSSI) to enhance resistance against spoofing attacks. Unsupervised learning is best applied in anomaly detection, where the system can find unusual data points that may signal a security threat. Deep learning models, including convolutional neural networks (CNNs) and recurrent neural networks (RNNs), improve pattern analysis to enhance detection in more advanced cyber-attacks [11].

The effective deployment of ML and DL in IoT security requires addressing challenges such as the need for large datasets and computation limitations presented by IoT devices. To adhere to data protection regulations, privacy and ethical considerations must also be considered. Further exploration is recommended in federated learning for decentralized data processing and federated learning to integrate blockchain technology with IoT to secure the framework [12]. While many approaches have proposed strategies for assuring data integrity and trust among devices, integration with blockchain technology offers a new frontier that can provide an alternative solution. Deepak et al. focused on using blockchain to develop a decentralized security framework for IoT networks [13]. They aimed to minimize the dependency on centralized data storage and reduce the corresponding single points of failure. Mazhar et al. [14] explores the increasing cyberattack vectors that threaten intelligent IoT applications, emphasizing the need for IoT systems to evolve with AI-centric ML and DL methodologies. This research underscores the utility of ML and DL in extracting actionable intelligence from raw data, crucial for fortifying IoT devices against diverse cyber threats. Additionally, federated learning models present an innovative solution for overcoming data privacy and ethical challenges inherent in traditional cloud-based ML approaches. Researchers in [15] highlight federated learning (FL) and DL as viable solutions for enhancing IoT security, emphasizing privacy-preserving data sharing as a superior alternative to conventional ML approaches.

Moreover, the work in [16] significantly advances IDS in IoT security. They propose a novel framework using few-shot learning (FSL) to enhance IDS efficacy in IoT environments. Their study introduces an innovative intrusion detection model combining a CNN for feature extraction with a prototypical network for FSL classification, achieving an accuracy of 99.44 % in detecting intrusions with minimal training data. Finally, Alshamrani's comprehensive survey on deploying IoT and AI for remote healthcare monitoring (RHM) systems within smart city frameworks highlights the critical intersection of technology and healthcare. This study outlines the technologies, devices and systems that underpin RHM systems based on IoT, highlighting the synergy between AI, ML, and IoT technologies in the collection and analysis of health-related data [17]. This insight into IoT and AI implementation in remote healthcare provides a valuable perspective on the broader implications of these technologies for IoT security, particularly in safeguarding sensitive health data and ensuring the integrity of healthcare delivery systems. The challenges highlighted, particularly those associated with deploying complex ML models within resource-constrained IoT environments and the pressing need for robust data protection measures, emphasize the critical nature of the forthcoming explorations. This research addresses these challenges by proposing a novel solution that combines advanced detection techniques based on XGBoost and LGBM classifiers, ensuring effective identification and response to modern threats.

3. Materials and methods

The proposed framework employs a systematic approach to assess the effectiveness of ensemble classifiers in detecting cyber threats across

IoT networks, framework architecture as shown in Fig. 1. It encompasses various stages, including data gathering, preprocessing, exploratory data analysis, model training and evaluation, and statistical analysis. This section outlines a rigorous, systematic methodology designed to enhance cyber threat detection within IoT networks, highlighting the novelty and comprehensive nature of the techniques used. A central aspect of this research is the use of the RT-IoT2022 dataset [10], which consists of authentic IoT traffic and simulated cyber-attacks, capturing the complexity and diversity of the IoT ecosystem. The selection of this dataset is a strategic choice aimed at basing the study on scenarios that accurately represent the complex nature of cyber threats prevalent in today's IoT networks. Furthermore, implementing sophisticated ML algorithms, including SMOTE [18] to address class imbalances and a random forest (RF) classifier to analyze the importance of characteristics, further differentiates this study. These methods are well-regarded for their ability to manage high-dimensional data and their resistance to overfitting and are employed meticulously to unravel the complexities of IoT cyber threats. The investigation extends into a detailed analysis of influential features, highlighting the intricate dynamics of network traffic and its vulnerability to IoT attacks.

3.1. RT-IoT2022 dataset

The RT-IoT2022 dataset serves as the cornerstone of this research, providing a rich source of real-world IoT traffic and simulated cyber-attacks. The dataset includes data from various IoT devices and attack scenarios, meticulously captured to reflect the complexity of real-time network interactions. Data preprocessing involved cleaning, normalization, and feature selection to prepare the dataset for analysis. This

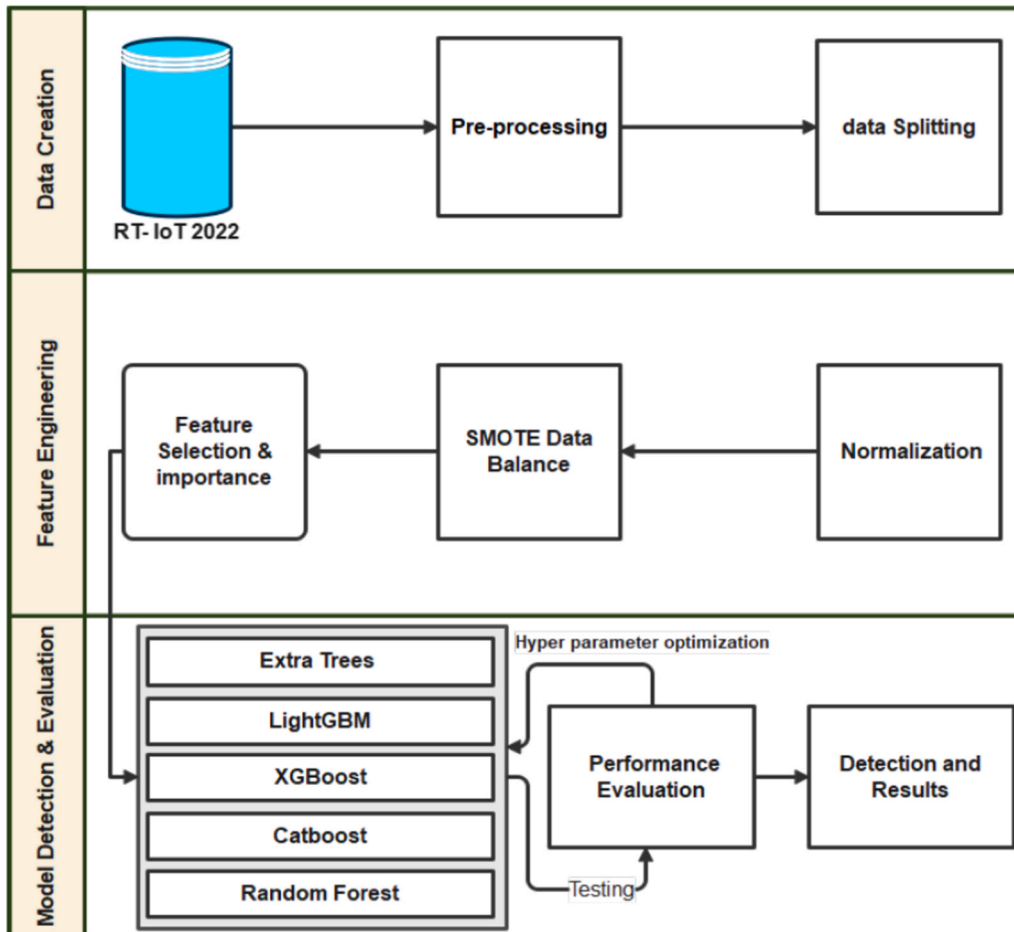


Fig. 1. Basic architectural block diagram of the proposed RT-IoT data modeling framework.

process aimed to enhance the quality of the data, ensuring that the subsequent analysis would yield reliable and accurate results. It is a proprietary compilation derived from real-time monitoring of IoT infrastructure. It is designed to reflect the diverse ecosystem of IoT devices and the myriads of potential network attack methodologies. By integrating data from a range of IoT devices, including but not limited to ThingSpeak-LED, Wipro-Bulb, and MQTT-Temp, alongside simulated scenarios of common cyber-attacks such as Brute-Force SSH attacks, DDoS attacks via Hping, and network mapper (Nmap) scanning patterns, the dataset offers a rich, multi-faceted view of the operational dynamics within IoT networks. This holistic approach aims to equip researchers and cybersecurity professionals with the insights needed to understand and counteract the evolving threats within these networks.

3.2. SMOTE based data balancing

The Class imbalance poses a significant challenge in cybersecurity datasets, where certain attack types are much less frequent than others. This discrepancy can cause a predictive model to favor the majority class, thereby reducing its effectiveness in identifying minority classes. To mitigate this imbalance, we implemented the SMOTE algorithm [18]. SMOTE addresses this issue by generating synthetic examples of the minority class, thus achieving a more balanced class distribution. Specifically, for a given minority class sample x_i , SMOTE identifies one of its k nearest neighbors, x_{zi} , and generates a new sample x_{synth} using the formula:

$$x_{synth} = x_i + \phi \times (x_{zi} - x_i) \quad (1)$$

Where ϕ is a randomly chosen fraction between 0 and 1, this parameter controls the degree of interpolation between the selected samples, facilitating the generation of realistic synthetic samples contributing to class balance. Upon addressing the imbalance with SMOTE, we utilized an RF classifier with default implementation [19] to discern the most informative features for classifying IoT attack types. The RF classifier is particularly advantageous for its ability to handle complex, high-dimensional datasets and its inherent protection against overfitting. The model computes the importance of each feature based on its contribution to accuracy gains within the decision trees, averaged over the entire ensemble of trees.

3.3. Comprehensive identification of influential features

The basic analysis using an RF classifier has identified key features that accurately identify types of IoT attacks. These features have varying degrees of importance, which are quantified below:

- `id.resp_p` (0.0595): Signifies the destination port number, a strong indicator of targeted services during an attack.
- `fwd_last_window_size` (0.0420): Reflects the size of the last window advertised by the sender, shedding light on the potential volume of an attack.
- `fwd_header_size_min` (0.0388): The smallest header size observed in the forward direction is relevant for packet-based attack detection.
- `flow_pkts_payload.max` (0.0365): Maximum payload size across all packets in a flow, critical for identifying large-scale data transmissions characteristic of certain attacks.
- `fwd_PSH_flag_count` (0.0355): Count of the PUSH flag in forward packets, useful for detecting urgent data indicative of attack patterns.
- `fwd_URG_flag_count` (0.0347): Count of urgent packets, which can signal attempts to prioritize malicious data.
- `flow_iat.min` (0.0337): Minimum inter-arrival time of packets, helping to identify rapid packet transmissions.
- `flow_FIN_flag_count` (0.0327): The number of FIN flags seen in a session could indicate session termination patterns specific to certain attacks.

- `fwd_pkts_payload.avg` (0.0320): The average payload size in the forward direction, highlighting typical data volumes in benign versus malicious traffic.
- Additional features such as `flow_pkts_payload.avg`, `fwd_init_window_size`, and `fwd_header_size_tot`, among others, underscore the intricate relationship between various network traffic attributes and malicious activities.

The selected features demonstrate the complex nature of network traffic and how it is used in attacks on IoT devices. We can detect a wide range of attack vectors by analyzing these features in detail. This analysis gives us better understanding how the predictive model works and helps us develop more advanced security measures to combat sophisticated cybersecurity threats. A series of visualizations were created to illustrate the significance of the identified features. These visual representations provide invaluable insights into the dynamics of the selected features and highlight their pivotal roles and interactions in the context of IoT security. The bar chart, represented in Fig. 2, visually underscores the weight of each feature, confirming that attributes such as `id.resp_p` and `fwd_last_window_size` are integral to the model's predictive power. Notably, the `id.resp_p` feature exhibits significant variability across attack types, as illustrated in the corresponding plot in Fig. 3, suggesting its capacity to distinguish between various attack strategies. The distribution of `fwd_last_window_size`, shown in Fig. 4, highlights its distinct patterns when comparing normal traffic to attack scenarios, emphasizing its role in characterizing the nature of the network behavior.

The significance of the feature `fwd_header_size_min` in the cybersecurity landscape is evident in Fig. 5. This feature's variability across different types of attacks highlights its crucial role in identifying anomalies in packet structures that could indicate malicious activities. The box plot helps visualize the distribution and range of the minimum header size in the forward direction of the traffic. This metric could signal the sophistication or type of an attack, particularly when considering protocol-specific vulnerabilities.

Fig. 6 delves into the `flow_pkts_payload.max` attribute, analyzing its distribution across various attack scenarios. This maximum payload size across all packets in a flow is a critical indicator of data volume anomalies, which could point to large-scale data exfiltration or flood attacks, characteristic of DDoS strategies. The analysis elucidates the role of payload size in differentiating between benign and malicious network activities, providing an additional dimension for cyber threat recognition and classification. After establishing the importance of selected features, box plots were meticulously crafted for the top four features, aiming to examine their distribution across different attack types. The correlation heatmap in Fig. 7 demystifies the interplay among the top 5 features. High correlation coefficients suggest a strong relationship, indicating coordinated attack mechanisms or a shared attack infrastructure. This visualization unravels the intricate relationships between these features, shedding light on their collective influence on attack classification.

The comprehensive visual analysis facilitated by these figures significantly enhances the comprehension of how selected features function in synergy to bolster the predictive model's efficacy. It underscores the importance of meticulous feature selection and underscores the complexities inherent in IoT security, aiding in the advancement of robust defensive strategies. The detailed commentary on these features, supported by visual evidence, enriches the understanding of the data's narrative. It provides a clear link between the abstract notion of feature importance and its practical implications in the real world, underlining the intricate nuances of network traffic behavior in the face of cybersecurity threats. Through this exposition, we offer a robust foundation for detecting a wide spectrum of attack vectors, thereby augmenting the development of advanced security measures. In the domain of machine learning, the choice and utilization of algorithms play a crucial role in developing models that are both accurate and computationally efficient. Moreover, they should be easy

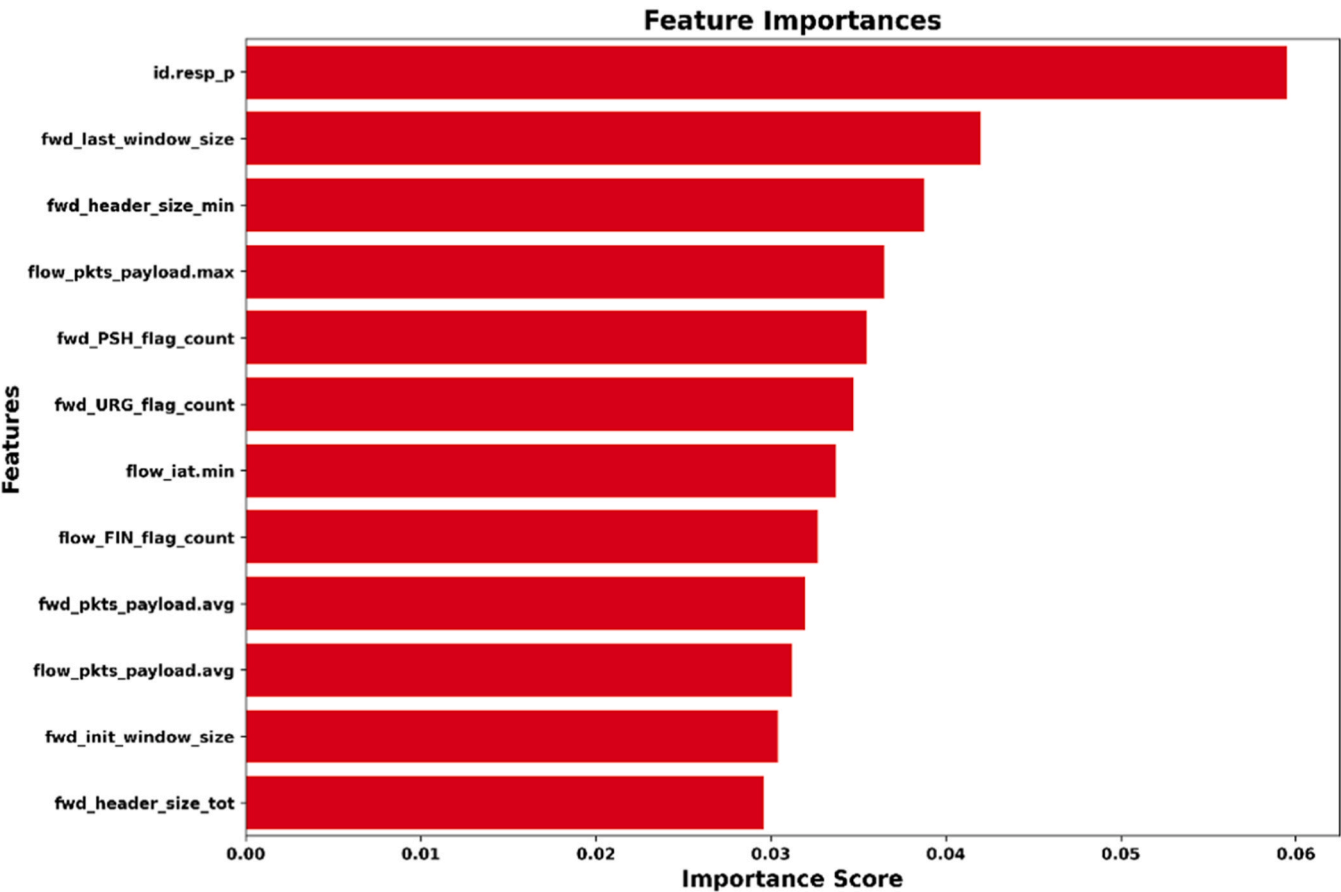


Fig. 2. Bar chart illustrates the importance scores of selected features, emphasizing the critical attributes that contribute significantly to the predictive model.

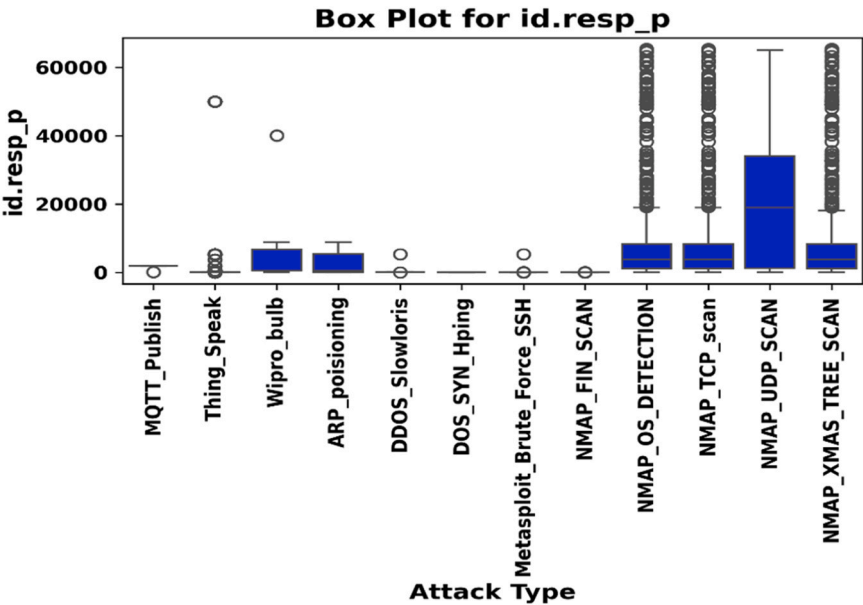


Fig. 3. Distribution of the id.resp_p feature across different attack types, showcasing its variability and the potential for distinguishing between attack categories.

to interpret. In this regard, we will discuss fundamental algorithms renowned for their unique approach to solving classification and prediction problems. We will also delve into Extra Trees, which boast an intuitive decision-making process and the ability to visualize complex decision paths. We will tailor the discussion of each algorithm to highlight its optimal usage scenarios, strengths, and limitations, particularly in dealing with complex cybersecurity datasets where class imbalance and the need for accurate predictions is of utmost importance.

3.4. CatBoost classifier

The categorical boosting (CatBoost) classifier [20] represents a

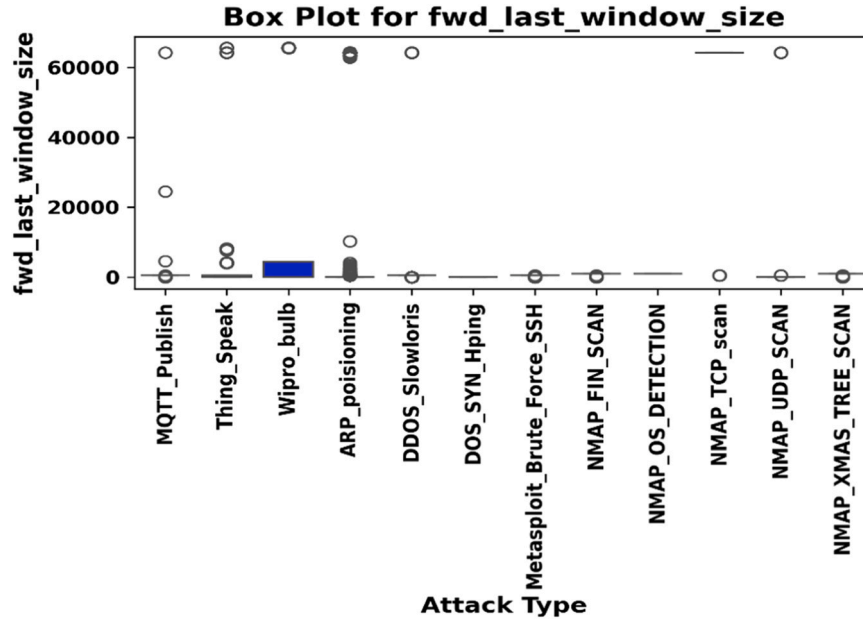


Fig. 4. Distribution of the fwd_last_window_size feature, illustrating its discriminative power in identifying specific attack vectors.

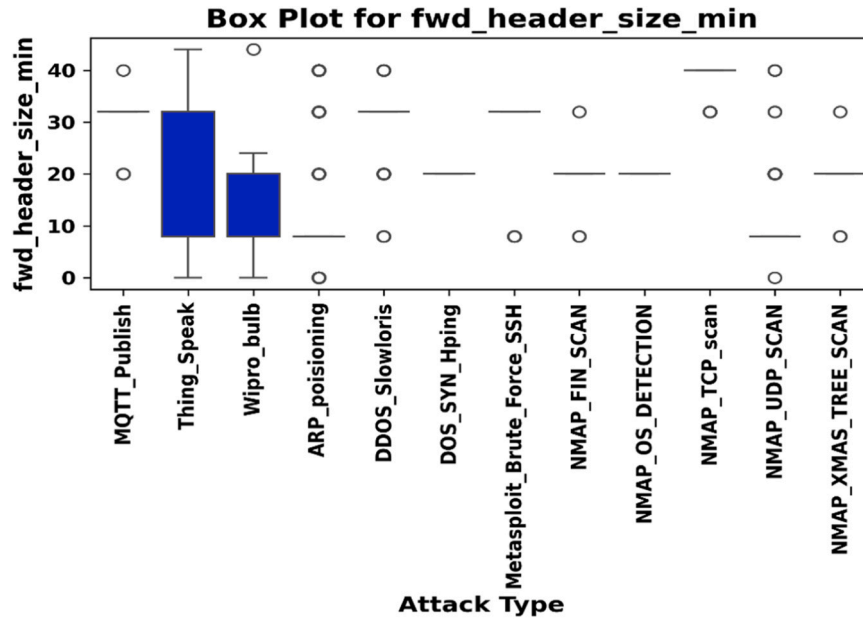


Fig. 5. Variability of the fwd_header_size_min feature across different attack scenarios, highlighting its importance in the context of cyber threat detection.

sophisticated evolution in the realm of machine learning algorithms, specifically designed to enhance predictive accuracy by efficiently handling categorical variables within its gradient boosting framework. Developed by Yandex, CatBoost automates the process of category encoding, which significantly reduces the preprocessing steps and mitigates common issues like overfitting. Central to the CatBoost approach is the implementation of ordered boosting, a permutation-driven technique that decreases the likelihood of overfitting and provides more stable and accurate predictions:

$$\min_{\theta} \sum_{i=1}^n L(y_i, \hat{y}_i) + \lambda \sum_{k=1}^K \Omega(f_k) \quad (2)$$

Here, $L(y_i, \hat{y}_i)$ denotes the loss function comparing the actual labels y_i to the predictions $\hat{y}_i$. At the same time, $\Omega(f_k)$ represents the regularization of the k^{th} model, with λ being a regularization parameter that controls

model complexity to prevent overfitting. Θ encapsulates the ensemble parameters. This formulation highlights CatBoost's commitment to delivering robust predictions while effectively managing categorical data complexities. A distinctive feature of CatBoost is its innovative handling of categorical data. It combines one-hot encoding for features with a few categories. It means encoding for more complex categories, which is integrated during training to avoid leakage and ensure model integrity. This method allows CatBoost to process categorical data directly without extensive data preprocessing, maintaining data purity and enhancing model performance.

The performance of the CatBoost classifier is enhanced by its ability to use symmetric trees. This optimization leads to faster training times than other boosting algorithms, making CatBoost ideal for large-scale applications and complex datasets in the finance, healthcare, and e-commerce sectors. CatBoost can be applied to various use cases,

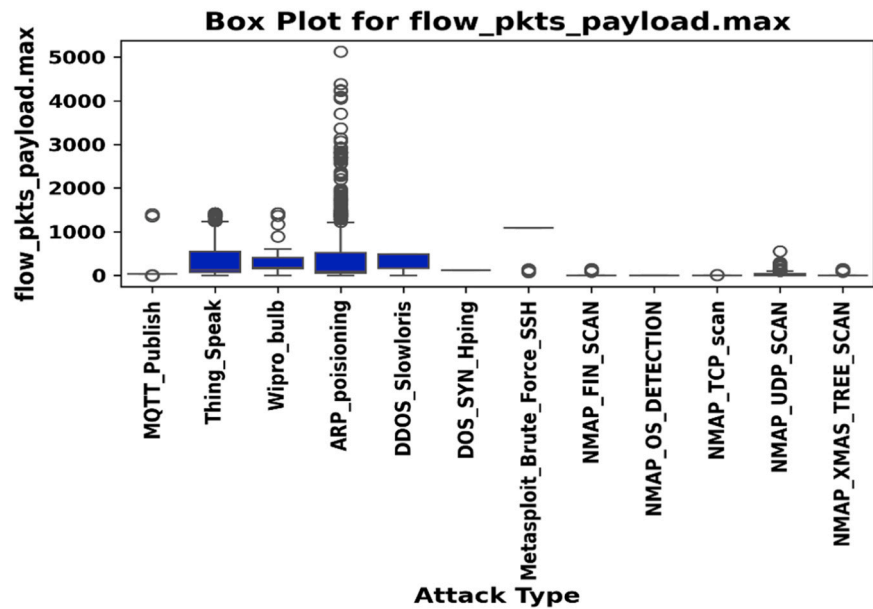


Fig. 6. Analysis of the flow_pkts_payload.max feature, demonstrating its role in distinguishing between benign and malicious network activities.

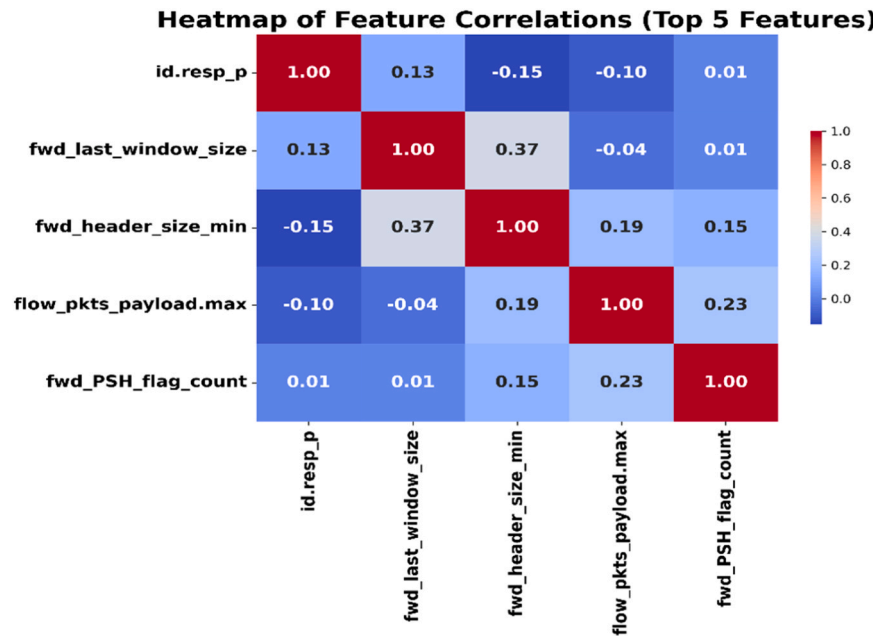


Fig. 7. Heatmap showing the correlation matrix for the top five features, revealing insights into how these features interrelate and their collective impact on attack classification.

including risk assessment, predictive maintenance, recommendation systems, and user behavior prediction. Its ability to handle large datasets with numerous categorical features and minimal preprocessing makes it an excellent choice for industries with high dimensional data. Traditional models might need extensive feature engineering, but CatBoost’s built-in handling of categorical variables and automated encoding make it a powerful tool. This capability is beneficial for complex user demographics and transactional data scenarios. It enables businesses to derive deep insights and predictive accuracy, which drives decision-making processes. The CatBoost classifier is a critical resource in predictive modeling, mainly where categorical data plays a significant role. Its robust design, innovative handling of categorical variables, and computational efficiency position CatBoost as a powerful tool for modern data scientists and analysts looking to leverage machine learning for

real-world problem-solving.

3.5. Extra trees classifier

The extremely randomized trees (Extra Trees) classifier [21] is an important component of ensemble learning. It is known for constructing many decision trees and combining their results to enhance the accuracy of predictions and reduce overfitting. Extra Trees is an extension of the RF algorithm, which introduces more randomness into the model building process. This can result in faster training times and a stronger, more robust model. The Extra Trees algorithm randomly selects data points and features at each split, instead of searching for the most optimal split points like traditional decision tree methods.

$$E(\Theta) = \frac{1}{N} \sum_{i=1}^N L(y_i, f(x_i; \Theta)) \quad (3)$$

where $E(\Theta)$ represents the expected error over all trees, $L(y_i, f(x_i; \Theta))$ denotes the loss function assessing the difference between actual labels y_i and predictions from tree f parameterized by Θ , and N is the number of trees. This equation underscores the aim of Extra Trees to minimize the average prediction error across all generated trees, leveraging the diversity introduced by its randomization technique.

One of the unique aspects of the Extra Trees classifier is its method of using the whole learning sample to grow the trees and randomly splitting nodes which contrasts with the more common strategies of bootstrapping samples and searching for the best splits used in other ensemble methods such as RF. The proposal only reduces the variance of the model without a substantial increase in bias but also significantly decreases the computational burden, making it highly efficient, especially in large datasets. Due to its efficiency and effectiveness, the Extra Trees classifier is widely used in various fields requiring robust predictive capabilities. It is particularly beneficial in complex regression and classification problems across bioinformatics, financial forecasting, and environmental modeling, where the intricate relationship between variables and data sets is substantial. The ability to handle high-dimensional data makes Extra Trees an excellent choice for image recognition tasks, where it can discern subtle patterns and variations that simpler models might overlook. Moreover, its inherent randomness helps provide a natural resistance to model overfitting, making it ideal for scenarios where predictive accuracy is crucial without the extensive need for parameter tuning. The Extra Trees classifier provides a powerful, efficient, and versatile tool in the arsenal of machine learning techniques. By integrating complete randomness in tree construction and combining the predictions of many such trees, Extra Trees delivers high-performance models that are both simple to implement and capable of handling a wide array of complex tasks.

3.6. Random Forest classifier

The RF classifier [22] is a powerful ensemble learning technique that combines multiple decision trees to produce a more accurate and stable prediction than any single tree could offer. This approach mitigates the common issues of overfitting associated with decision trees, particularly in complex datasets with numerous input variables. The foundation of the RF algorithm lies in the creation of an ensemble of decision trees, each constructed from a random subset of the training data and using a random selection of features at each split in the tree.

$$RF(\Theta) = \frac{1}{M} \sum_{m=1}^M L(y, f(x; \Theta_m)) \quad (4)$$

where $RF(\Theta)$ denotes the RF model, M is the number of trees in the forest, $L(y, f(x; \Theta_m))$ represents the loss function assessing the prediction errors from the m^{th} tree, and Θ_m are the parameters defining each tree. This formula highlights the algorithm's goal to average the predictions across a forest of trees, reducing the variance without increasing bias significantly. The success of RF lies in its bagging strategy, where each tree is trained on different data samples while performing feature selection randomly. It creates diversity among the trees, resulting in an overall improvement in predictive performance and model robustness. As a result, RF has become a popular classifier in various domains because of its versatility, ease of use, and robust performance in classification and regression tasks. It is widely used in fields such as biology for genomic classification and medical diagnosis, finance for credit scoring and fraud detection, and e-commerce for predicting customer behavior and inventory classification. RF is a robust machine learning algorithm with many strengths. One of its key features is its ability to evaluate the importance of predictors, helping analysts identify which variables have the most impact on forecasting outcomes. It is beneficial in exploratory studies where understanding the impact of variables is

crucial. Moreover, RF can handle large datasets with high dimensionality and provide estimates of important variables in classification. It also effectively estimates missing data and maintains accuracy even when a significant proportion is missing. RF is an essential machine learning component, offering a reliable, flexible, and easy-to-use approach that can be applied to a wide range of practical problems. It is particularly attractive for academic research and industrial applications because it can operate accurately even with minimal tuning.

3.7. XGBoost classifier

The eXtreme gradient boosting (XGBoost) classifier [23] emerges as an advanced iteration within ensemble learning, which focuses on amplifying predictive precision by aggregating more straightforward model outputs. It employs a refined gradient boosting framework to iteratively enhance models by addressing errors identified in preceding iterations. Central to XGBoost's approach is the optimization of a regularized objective:

$$\min_{\Theta} \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (5)$$

Here, $L(y_i, \hat{y}_i)$ denotes the loss between actual labels y_i and predictions $\hat{y}_i$, while $\Omega(f_k)$ represents the regularization of the k^{th} model, aimed at mitigating overfitting by penalizing model complexity, with Θ indicating the ensemble parameters. This formulation underscores XGBoost's dual focus on predictive accuracy and model parsimony. XGBoost's acclaim stems from its computational efficiency, scalability, and robustness against overfitting, courtesy of its regularization mechanism. The algorithm's capacity to manage missing data and its system optimizations contribute to its enhanced speed and performance relative to conventional boosting methods. However, the need for meticulous hyperparameter tuning and the computational demands of processing extensive data sets present notable challenges. XGBoost's versatility is showcased through its applications in various sectors, from achieving unparalleled accuracy in competitive ML arenas to practical deployments in customer analytics, fraud prevention, and recommendation systems, marking it as an indispensable tool in predictive modeling.

4. Experimental design and setup

This study evaluates the effectiveness of various ML algorithms in identifying and categorizing cybersecurity threats in IoT environments. This section describes the used dataset, preprocessing techniques, algorithm selection process, evaluation metrics, and experimental environment setup. We have used a comprehensive IoT cybersecurity dataset, including benign and malicious network traffic patterns. The dataset contains a diverse range of attack types labeled for supervised learning, such as distributed denial of service (DDoS), Man in the middle (MitM), and malware injections. It also includes several features, such as packet sizes, protocol types, and timestamps, which create a rich and multidimensional dataset ideal for in-depth analysis. Data preprocessing is a critical step that ensures the quality and compatibility of the dataset with the learning algorithms. The following steps are including:

1. Data Cleaning: Remove duplicate entries and handle missing values through imputation techniques, enhancing dataset integrity.
2. Feature Engineering: Extraction and construction of informative features from raw data, alongside eliminating redundant features, as informed by the feature importance analysis detailed in Section 3.3.
3. Normalization: Application of Min-Max normalization to scale the feature values, facilitating algorithm convergence and performance stability.
4. Class Balancing: Implementation of SMOTE to address class imbalance, as elaborated in Section 3.2, ensuring equitable class representation.

In this study, we selected CatBoost, Extra Trees, RF, XGBoost, and LightGBM, each bringing specific strengths that can be effectively leveraged in an ensemble strategy. To enhance the performance of these models, we implement a grid search for automated parameter tuning. Table 1 details each algorithm's hyperparameter optimization and highlighted to the automated selected parameters for each classifier. It ensures that each classifier is finely tuned to adjust each model's configuration to balance predictive performance and computational efficiency.

The experiments took place in a controlled environment, utilizing a top-of-the-line workstation with an Intel Core i7 processor, 32 GB RAM, and an NVIDIA GTX 1080 Ti GPU. We used Python 3.8, Scikit-learn, and TensorFlow to handle computationally intensive tasks such as the SMOTE application and XGBoost training.

5. Results and findings analysis

This section presents a detailed analysis of the performance of various classifiers employed to detect and classify cybersecurity threats within IoT environments.

Table 2 provides the comparative performance metrics of some advanced machine learning classifiers in identifying cyber security threats. In this analysis, CatBoost performs well for all the folds averaging an accuracy ratio of 98.85 %. Extra Trees produces promising results, but its average accuracy is only 95.10 %. It highlights its level of relevance in this case. RF achieves a remarkable average of 98.28 %, showcasing its effectiveness in threat detection. The performance of

Table 1

An automated parameter tuning for each classifier using a grid search algorithm.

Classifier	Hyperparameters	Grid Search Parameters
CatBoost	iterations (Number of boosting rounds)	[100, 300, 500]
	learning_rate (Step size for each iteration)	[0.01, 0.05, 0.1 , 0.2]
	depth (Maximum depth of trees)	[6, 8 , 10]
	l2_leaf_reg (L2 regularization)	[3, 5, 7]
	border_count (Number of splits for features)	[32, 64 , 128]
Extra Trees	n_estimators (Number of trees in the forest)	[100, 200 , 500]
	max_depth (Maximum depth of trees)	[None, 10, 20, 30]
	min_samples_split (Min samples to split a node)	[2, 5, 10]
	min_samples_leaf (Min samples in a leaf node)	[1, 2, 4]
	max_features (Number of features to consider)	['auto', ' sqrt ', 'log2']
RF	n_estimators (Number of trees in the forest)	[100, 200 , 500]
	max_depth (Maximum depth of trees)	[None, 10, 20, 30]
	min_samples_split (Min samples to split a node)	[2, 5, 10]
	min_samples_leaf (Min samples in a leaf node)	[1, 2, 4]
	max_features (Number of features to consider)	['auto', ' sqrt ', 'log2']
XGBoost	n_estimators (Number of boosting rounds)	[100, 300, 500]
	learning_rate (Step size for each iteration)	[0.01, 0.05, 0.1 , 0.2]
	max_depth (Maximum depth of trees)	[6, 8 , 10]
	subsample (Fraction of samples for training)	[0.6, 0.8 , 1.0]
	colsample_bytree (Fraction of features per tree)	[0.6, 0.8 , 1.0]
LightGBM	n_estimators (Number of boosting rounds)	[100, 300 , 500]
	learning_rate (Step size for each iteration)	[0.01, 0.05, 0.1 , 0.2]
	max_depth (Maximum depth of trees)	[-1, 10 , 20, 30]
	num_leaves (Max number of leaves in a tree)	[31, 61, 127 , 255]
	subsample (Fraction of samples for training)	[0.6, 0.8 , 1.0]
	colsample_bytree (Fraction of features per tree)	[0.6, 0.8 , 1.0]

*Bold values indicate the best parameters.

Table 2

Comparative performance metrics of advanced ML classifiers in cybersecurity threat detection.

Fold	CatBoost	Extra Trees	RF	XGBoost	LightGBM
Fold 1	98.74	95.14	98.31	99.52	99.65
Fold 2	98.88	95.20	98.32	99.55	99.64
Fold 3	98.97	94.88	98.23	99.53	99.66
Fold 4	98.85	95.12	98.24	99.54	99.65
Fold 5	98.82	94.99	98.30	99.56	99.63
Fold 6	98.95	95.05	98.26	99.55	99.65
Fold 7	98.77	95.10	98.24	99.54	99.64
Fold 8	98.79	95.15	98.25	99.56	99.66
Fold 9	98.82	95.18	98.28	99.54	99.65
Fold 10	98.86	95.14	98.32	99.65	99.66
Avg CV Accuracy	98.85	95.10	98.28	99.55	99.65

XGBoost is above average, with its accuracy at 99.55 %, implying that it is able to withstand different attacks. LightGBM also performs well with an accuracy average of 99.65 %. It indicates its suitability in this area. The results have shown that there is performance of good strength by all classifiers. However, XGBoost and LightGBM are better performers, making these models a better fit for network IDSs to secure the IoT.

Classifiers evaluated include CatBoost, Extra Trees, RF, XGBoost, and LGB, with their performance metrics summarized in Table 3. The classifiers demonstrate varied levels of effectiveness across different metrics. Notably, XGBoost and LGBM exhibit superior performance, with accuracy scores exceeding 99 %, closely followed by RF. CatBoost, while competitive, shows a slight performance decrease relative to these leaders. Despite its robustness and computational efficiency, Extra Trees lags in terms of precision and recall. The high accuracy, precision, and recall scores of XGBoost and LGBM suggest their exceptional capability in correctly identifying and classifying most attack instances without significant overfitting, as evidenced by their F1-Scores and Matthew's correlation coefficient (MCC). The Cohen Kappa score further underscores the substantial agreement between the predictions and the actual labels, beyond what would be expected by chance. The relatively lower scores of Extra Trees, in comparison, highlight potential limitations in handling the dataset's complexity or a need for more fine-tuned hyperparameters.

The performance distinctions among classifiers underscore the importance of algorithm selection in cybersecurity applications. The superior metrics of XGBoost and LGBM can be attributed to their advanced ensemble strategies that effectively mitigate overfitting while capturing complex, non-linear relationships within the data. Their efficacy is particularly notable in high-dimensional spaces common in cybersecurity datasets, where discerning subtle patterns is critical for accurate threat detection and classification. Also, to assess the performance differences among various machine learning classifiers in detecting cybersecurity threats, paired t-tests on the F1-Scores obtained from classifiers is conducted, focusing particularly on XGBoost and LGBM because of their noteworthy performance. The established

Table 3

Comparative performance metrics of advanced ML classifiers in cybersecurity threat detection.

Metric	CatBoost	Extra Trees	RF	XGBoost	LGBM
Avg CV	98.847	95.102	98.278	99.553	99.651
Precision	98.88	93.141	98.245	99.551	99.648
Recall	98.847	95.102	98.278	99.553	99.651
F1-Score	98.743	93.653	98.021	99.549	99.647
Cohen Kappa	97.129	87.045	95.713	98.889	99.131
Score					
MCC	97.132	87.404	95.732	98.889	99.132
Zero-One Loss	0.0115	0.049	0.017	0.0045	0.0035
Hamming Loss	0.0115	0.049	0.017	0.0045	0.0035

*The best results are highlighted in Bold.

significance level for these tests was $\alpha = 0.05$, following standard statistical conventions.

The statistical analysis was structured around two hypotheses for each comparison:

- Null Hypothesis (H_0): No significant difference exists in the mean F1-Scores between the classifier pairs.
- Alternative Hypothesis (H_1): A significant difference is present in the mean F1-Scores between the classifier pairs.

A detailed examination of the paired *t*-test results revealed the following findings:

- A significant discrepancy between XGBoost and CatBoost, as evidenced by a p-value of 0.0012, leading to H_0 's rejection.
- The XGBoost vs. Extra Trees comparison, with a p-value of 2.00e-07, confirmed a significant performance gap, dismissing H_0 .
- XGBoost's comparison with RF resulted in a p-value of 5.33e-06, indicating a notable difference and the rejection of H_0 .
- The match-up between XGBoost and LGBM yielded a p-value of 0.0446, signifying a significant, albeit marginal, difference, suggesting their comparable prowess.
- The *t*-test between LGBM and CatBoost produced a p-value of 1.29e-05, underscoring LGBM's superiority and leading to H_0 's rejection.
- LGBM's comparison with Extra Trees, revealing a p-value of 3.12e-07, and with RF, showcasing a p-value of 7.57e-07, both illustrated significant performance differences favoring LGBM.

The statistical analysis revealed the performance hierarchy among classifiers, with XGBoost and LGBM models emerging as the most dominant. Although these two models exhibited nearly equivalent effectiveness in classifying cybersecurity threats, their nuanced yet statistically significant differences suggest that they can be used interchangeably based on context specific requirements. It underscores the strategic advantage of employing advanced ensemble techniques in complex cybersecurity landscapes. These findings validate the exceptional capabilities of XGBoost and LGBM, which are facilitated by gradient boosting methodologies. They also highlight the importance of statistical validation in substantiating classifier selections. The ability of these algorithms to parse through data and identify threats further affirms their critical utility in crafting precision-oriented, reliable cybersecurity defenses. This subsection synthesizes a comprehensive statistical examination, accentuating significant, quantifiable distinctions in classifier performances. It underscores the indispensable value of meticulous algorithmic selection, reinforced by empirical evidence, in the ongoing endeavor to fortify cybersecurity frameworks. The formation of XGBoost and LGBM classifiers demonstrated remarkable success in identifying and neutralizing cyber threats, reinforcing the current research's tangible applicability and effectiveness in enhancing IoT security measures. The adaptability and precision of ML-based IDSs advocate for a proactive shift towards intelligent, data-driven security solutions, tailoring defenses to the nuanced threats IoT ecosystems face. The promising results validate the effectiveness of the XGBoost and LGBM models and pave the way for future advancements in IoT security frameworks, ensuring a safer, more secure digital environment for IoT devices.

6. Discussion

The study conducted has shown that ML algorithms can significantly enhance the security of IoT networks. The research leveraged the RT-IoT2022 dataset to highlight how network traffic performs in different operational and attack scenarios and to identify vulnerabilities within IoT ecosystems. The study compared ensemble classifiers, specifically XGBoost and LGBM, with other advanced ML algorithms and found that these classifiers are exceptional in detecting and classifying a wide range of cyber-attacks. Their superiority is because of their complexity and ability to identify subtle, nonlinear patterns within high-dimensional

data spaces, showing the evolution of ML in cybersecurity. The study concludes that IoT security measures should be augmented with AI-driven IDSs. It also underscores that selecting an appropriate algorithm is crucial to its performance and is specific to network characteristics and threat profiles. The study has significantly contributed to the academic and practical discourse on IoT security. Firstly, it validates the RT-IoT2022 dataset as a pivotal resource for developing and benchmarking IDS solutions. Secondly, the rigorous statistical analysis undertaken to compare classifier performances reinforces the validity of the experimental findings and introduces a methodological rigor often sought in cybersecurity research. The study highlights the need for IoT security solutions to be adaptive to the dynamic nature of cyber threats. The proposed framework for integrating advanced ensemble classifiers into real-world IoT networks offers a blueprint for enhancing resilience against cyber threats, paving the way for future innovations in the domain. However, the study acknowledges limitations, such as the computational demands of processing extensive datasets and the need for continuous model retraining to adapt to evolving threat patterns. The study also recognizes ethical considerations surrounding data privacy and the potential misuse of AI in cybersecurity, emphasizing the need for responsible AI governance and ethical AI frameworks.

7. Conclusion and future work

This research culminates in a comprehensive understanding of the challenges and opportunities within IoT security through meticulous analysis of IoT network traffic and evaluation of cutting-edge ML classifiers. The findings highlight critical vulnerabilities prevalent in IoT networks and demonstrate the transformative potential of ML and AI technologies in mitigating these risks. The study's primary contributions include a detailed exploration of the RT-IoT2022 dataset, revealing key insights into IoT network behaviors and threat dynamics. The benchmarking of advanced ensemble classifiers, particularly XGBoost and LGBM, establishes new standards in performance and efficacy for IDSs with accuracy rates of 99.553 % and 99.651 % respectively. Furthermore, statistical validation of these classifiers' superiority enhances the research's credibility and relevance. This study provides a robust foundation for further exploring hybrid ML models. Combining multiple classifiers can improve threat detection accuracy and efficiency.

Blockchain integration and federated learning models promise to enhance data security and privacy in IoT networks and this plan of intended future work using different datasets.

Ethical consent

The authors did not conduct any clinical studies or experiments on humans or animals.

CRedit authorship contribution statement

Muhammad Atta Othman Ahmed: Software, Conceptualization. **Omar Ahmed:** Writing – review & editing, Project administration, Methodology. **Raed Alotaibi:** Validation, Data curation. **Yasser AbdelSatar:** Writing – original draft, Formal analysis.

Declaration of Competing Interest

I declare that I have no significant competing financial, professional, or personal interests that might have influenced the performance or presentation of the work. Also, we do not have any conflict with the authors.

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